

Dang Nguyen

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Research interests

My research focuses on data-centric methods for building efficient and reliable agentic AI systems based on large (vision-)language models. I develop techniques for **synthetic data generation** and **data selection** to construct high-quality training signals under limited or noisy data. More recently, I have explored reasoning and decision-making, including **test-time scaling**, **RL training**, and **uncertainty-aware inference**, with the goal of enabling **reliable long-horizon agents** in high-stakes settings.

Education

University of California, Los Angeles

California, USA

Ph.D. in Computer Science

Sep. 2023 - Present

- Advisor: Professor Baharan Mirzasoleiman
- UCLA Graduate Dean's Scholar Award

Toyo University

Tokyo, Japan

B.S. in Information Networking for Innovation and Design

Apr. 2017 - Mar. 2021

- Toyo Top Global Scholarship A
- GPA: 4.27/4.3, Top 1/300 in the faculty
- Top thesis award

Experience

Adobe

California, USA

Applied Scientist Intern

Jun. 2026 - Sep. 2026

- Supervisors: Shubham Lohiya, Sizhe Zhou, Dr. Anay Majee
- Manager: Dr. Sajjadur Rahman
- Topic: Configurable Safety and Governance for Autonomous Agentic Systems
- Designed a runtime agent safety and governance system combining organization-level policies with a reasoning-blind bounded tool-call/result window to detect intent-action mismatches, tool-result prompt injection, and dangerous cross-tool chains; achieved 99.3% recall and 96.2% precision on 1k+ production approval requests, reducing human approval prompts by 95.5%.
- Developed a synthetic adversarial dataset generation and evaluation framework that automatically generates safety-critical agent trajectories through interactions with sandboxed backends and verifies their reachability and impact; constructed a benchmark exposing current failure modes and enabling systematic regression testing.
- Integrated the system into Adobe Experience Platform, enabling configurable runtime governance for autonomous enterprise agents.

Google Research

California, USA

Student Researcher

Sep. 2024 - Aug. 2025

- Topics: Synthetic data generation for LLMs and Data selection for LVLMS

Cisco

California, USA

PhD Research Intern

Jun. 2024 - Sep. 2024

- Supervisor: Dr. Ali Payani
- Topic: LLM Hallucination

VinAI (now Qualcomm AI)

AI Resident

Hanoi, Vietnam

Oct. 2020 - Aug. 2023

- Supervisor: Professor Nhat Ho (UT Austin)
- Topics: Optimal Transport and Model Merging
- Participated in an applied project which aims to improve the performance of object detectors in low-light conditions.
- Managed GPU resources for the VinAI Residency Program.

FPT Japan Holdings

Part-time Machine Learning Engineer

Yokohama, Japan

Oct. 2019 - Sep. 2020

- Participated in a long-term demand forecasting project for a chain pharmacy company in Japan.

Publications

(*) denotes equal contribution

1. D. Nguyen*, J. Huang*, A. Payani, B. Mirzasoleiman. Why is A+B Better Than B? A Simple Graph Perspective on Task Transfer. *ICML 2026 Workshop on Foundations of Deep Generative Models*.
2. N. Naharas*, **D. Nguyen***, N. Bulut, M. Bateni, V. Mirrokni, and B. Mirzasoleiman. Data Selection for Fine-tuning Vision Language Models via Cross Modal Alignment Trajectories. *International Conference on Machine Learning (ICML)*, 2026.
3. **D. Nguyen***, J. Li*, J. Zheng*, B. Mirzasoleiman. Do We Need All the Synthetic Data? Towards Targeted Synthetic Image Augmentation via Diffusion Models. *International Conference on Learning Representations (ICLR)*, 2026.
4. **D. Nguyen**, A. Payani, B. Mirzasoleiman. Beyond Semantic Entropy: Boosting LLM Uncertainty Quantification with Pairwise Semantic Similarity. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Findings)* 2025.
5. **D. Nguyen***, Z. Li*, M. Bateni, V. Mirrokni, M. Razaviyayn, and B. Mirzasoleiman. Synthetic Text Generation for Training Large Language Models via Gradient Matching. *International Conference on Machine Learning (ICML)*, 2025.
6. **D. Nguyen**, W. Yang, R. Anand, Y. Yang and B. Mirzasoleiman. Mini-batch Coresets for Memory-efficient Language Model Training on Data Mixtures. *International Conference on Learning Representations (ICLR)*, 2025.
7. **D. Nguyen**, P. Haddad, E. Gan, and B. Mirzasoleiman. Changing the Training Data Distribution to Reduce Simplicity Bias Improves In-distribution Generalization. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
8. Y. Xue, J. Siddharth, **D. Nguyen**, and B. Mirzasoleiman. Understanding the Robustness of Multi-modal Contrastive Learning to Distribution Shift. *International Conference on Learning Representations (ICLR)*, 2024.
9. K. Nguyen*, **D. Nguyen***, N. Ho. Self-Attention Amortized Distributional Projection Optimization for Sliced Wasserstein Point-Cloud Reconstruction. *International Conference on Machine Learning (ICML)*, 2023.
10. **D. Nguyen**, T. Nguyen, K. Nguyen, D. Phung, H. Bui, and N. Ho. On cross-layer alignment for model fusion of heterogeneous neural networks. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023. (Top 3%)
11. K. Nguyen*, **D. Nguyen***, T. A. V. Le, T. Pham, and N. Ho. Improving mini-batch optimal transport via partial transportation. *International Conference on Machine Learning (ICML)*, 2022.
12. K. Nguyen, **D. Nguyen**, Q. Nguyen, T. Pham, H. Bui, D. Phung, T. Le, and N. Ho. On transportation of mini-batches: A hierarchical approach. *International Conference on Machine Learning (ICML)*, 2022.

Honors & Awards

INTERNATIONAL

- 2026 **Amazon Fellows**, UCLA
2023 **UCLA Graduate Dean's Scholar Award**, UCLA
2017 **Toyo Top Global Scholarship A**, Toyo University
2015 **Silver medal**, 56th International Mathematical Olympiad

California, USA
California, USA
Tokyo, Japan
Chiang Mai, Thailand

DOMESTIC

- 2015 **First Prize**, Vietnam Mathematical Olympiad
2014 **Second Prize**, Vietnam Mathematical Olympiad

Hanoi, Vietnam
Hanoi, Vietnam

Professional services

- Reviewer at Conference on Neural Information Processing Systems (NeurIPS) 2022-2026
- Reviewer at the International Conference on Machine Learning (ICML) 2023-2026
- Reviewer at the International Conference on Learning Representations (ICLR) 2024-2026
- Reviewer at the International Conference on Artificial Intelligence and Statistics (AISTATS) 2023-2025
- Program Committee at AAAI 2025
- Program Committee at New Frontiers in AdvML@NeurIPS2024
- Reviewer at Workshop on Spurious Correlation and Shortcut Learning @ ICLR 2025